Chapter 9. The Fundamental Group

Path homotopy, covering spaces, the circle, retractions and fixed points, Borsuk–Ulam, homotopy type, spheres and surfaces

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Algebraic topology attaches algebraic objects to spaces so that topological questions (are two spaces homeomorphic? does a map have a fixed point?) become algebraic ones. The first such object is the fundamental group. Throughout, $I = [0, 1]$.

9.1 Homotopy of paths

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ with $F(x, 0) = f(x)$, $F(x, 1) = f'(x)$. Paths $f, f' : I \to X$ from x_0 to x_1 are **path homotopic**, $f \simeq_p f'$, if moreover $F(0, t) = x_0$ and $F(1, t) = x_1$ for all t. Both are equivalence relations; $[f]$ denotes the path-homotopy class.

Example 9.1

In a convex subset $A \subset \mathbb{R}^n$ any two paths with the same endpoints are path homotopic by the straight-line homotopy $F(s, t) = (1 - t)f(s) + tg(s)$.

If f ends where g begins, the **product** $f * g$ runs through f on $[0, \frac{1}{2}]$ and g on $[\frac{1}{2}, 1]$ (pasting lemma). It is well defined on classes: $[f] * [g] = [f * g]$.

Theorem 9.1 (Groupoid properties)

The operation $*$ on path-homotopy classes is associative (when defined), has identities $[e_x]$ (constant paths), and inverses $[\bar{f}]$, $\bar{f}(s) = f(1 - s)$.

Proof. The key observation: if $k : I \to I$ is continuous with $k(0) = 0$, $k(1) = 1$, then $f \circ k \simeq_p f$ (straight-line homotopy in I). Associativity: $(f * g) * h$ is a reparametrization of $f * (g * h)$. Identities similarly. For inverses, $F(s, t) = f(2s(1 - t))$ for $s \leq \frac{1}{2}$ and $f((2 - 2s)(1 - t))$ for $s \geq \frac{1}{2}$ contracts $f * \bar{f}$ to e_{x_0}. ∎

9.2 The fundamental group

Definition (Fundamental group)

For $x_0 \in X$, $\pi_1(X, x_0)$ is the set of path-homotopy classes of loops at x_0 with the operation $*$.

A path α from x_0 to x_1 induces an isomorphism $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$, $[f] \mapsto [\bar{\alpha}] * [f] * [\alpha]$. So for path connected X the group does not depend on the base point up to isomorphism (but the isomorphism depends on α unless π_1 is abelian). X is **simply connected** if path connected with trivial π_1; then any two paths with the same endpoints are path homotopic. Convex subsets of $\mathbb{R}^n$ are simply connected.

A continuous $h : (X, x_0) \to (Y, y_0)$ induces $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $[f] \mapsto [h \circ f]$, with $(k \circ h)_* = k_* \circ h_*$ and $(\mathrm{id})_* = \mathrm{id}$. Hence homeomorphisms induce isomorphisms: π_1 is a topological invariant.

9.3 Covering spaces

Definition (Covering map)

A continuous surjection $p : E \to B$ is a **covering map** if every $b \in B$ has an open neighbourhood U that is **evenly covered**: $p^{-1}(U)$ is a disjoint union of open sets V_α each mapped homeomorphically onto U by p.

Examples: $p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$; $z \mapsto z^n$ on S^1; products of covering maps such as $\mathbb{R}^2 \to S^1 \times S^1$. A covering map is a local homeomorphism and an open map; the converse fails ($\mathbb{R}_+ \to S^1$).

A **lifting** of $f : X \to B$ is $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

Lemma 9.2 (Path and homotopy lifting)

Let $p : (E, e_0) \to (B, b_0)$ be a covering. (a) Every path in B starting at b_0 has a unique lifting starting at e_0. (b) Every map $F : I \times I \to B$ with $F(0,0) = b_0$ has a unique lifting $\tilde{F}$ with $\tilde{F}(0,0) = e_0$; if F is a path homotopy, so is $\tilde{F}$.

Proof. (a) By the Lebesgue number lemma subdivide I so that each piece maps into an evenly covered set, and lift piece by piece using the local inverses of p. Uniqueness: two liftings agree on a set that is open and closed in I. (b) Do the same on a grid of small squares in lexicographic order; the edges $\{0\} \times I$ and $\{1\} \times I$ map to points, and their liftings are connected subsets of the discrete fibre, hence constant. ∎

Theorem 9.3 (Lifting correspondence)

Define $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$ by $\phi([f]) = \tilde{f}(1)$, the endpoint of the lift starting at e_0. It is well defined; if E is path connected it is surjective, and if E is simply connected it is bijective.

Proof. Well defined by homotopy lifting. Surjective: join e_0 to e_1 in E and push down. Injective: if $\tilde{f}(1) = \tilde{g}(1)$, then $\tilde{f} \simeq_p \tilde{g}$ in the simply connected E, and composing with p gives $f \simeq_p g$. ∎

9.4 The fundamental group of the circle

Theorem 9.4

$\pi_1(S^1, b_0) \cong \mathbb{Z}$.

Proof. Use $p : \mathbb{R} \to S^1$ with $e_0 = 0$, so $p^{-1}(b_0) = \mathbb{Z}$ and $\phi : \pi_1 \to \mathbb{Z}$ is a bijection ($\mathbb{R}$ is simply connected). It is a homomorphism: if $\tilde{f}$ ends at n and $\tilde{g}$ at m, then $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ and ends at $n + m$. ∎

More generally, for a covering $p : (E, e_0) \to (B, b_0)$, p_* is injective, the lifting correspondence induces a bijection from the right cosets of $H = p_*\pi_1(E, e_0)$ onto $p^{-1}(b_0)$, and a loop f lifts to a loop iff $[f] \in H$.

9.5 Retractions and fixed points

$A \subset X$ is a **retract** if there is a continuous $r : X \to A$ with $r|_A = \mathrm{id}$. Then $i_* : \pi_1(A) \to \pi_1(X)$ is injective (since $r_* i_* = \mathrm{id}$).

Theorem 9.5 (No-retraction theorem)

S^1 is not a retract of the closed disc B^2.

Proof. $\pi_1(S^1) = \mathbb{Z}$ cannot inject into $\pi_1(B^2) = 0$. ■

A loop $h : S^1 \to X$ is **nulhomotopic** iff it extends to B^2 iff h_* is trivial.

Theorem 9.6 (Brouwer fixed point theorem, $n = 2$)

Every continuous $f : B^2 \to B^2$ has a fixed point.

Proof. If $f(x) \neq x$ for all x, the ray from $f(x)$ through x meets S^1 in a point $r(x)$, and r is a retraction of B^2 onto S^1. ■

Corollary 9.7

A 3×3 matrix with positive entries has a positive eigenvalue (apply Brouwer to $x \mapsto Ax/\|Ax\|_1$ on the triangle $\{x \geq 0, \sum x_i = 1\}$). A nonvanishing vector field on B^2 points directly outward and directly inward at some boundary points.

9.6 The fundamental theorem of algebra and Borsuk–Ulam

Theorem 9.8 (Fundamental theorem of algebra)

A nonconstant complex polynomial has a root.

Proof. For monic $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ with $\sum|a_i| < 1$ (rescale z), the map $z \mapsto z^n$ on S^1 induces multiplication by n on π_1, hence is not nulhomotopic, and the homotopy $z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on S^1. If p had no root in B^2, $p|_{S^1}$ would extend to $B^2 \to \mathbb{R}^2 - 0$, hence be nulhomotopic in $\mathbb{R}^2 - 0$, a contradiction. ■

Theorem 9.9 (Borsuk–Ulam, $n = 2$)

For every continuous $f : S^2 \to \mathbb{R}^2$ there is x with $f(x) = f(-x)$. Equivalently there is no antipode-preserving map $S^2 \to S^1$.

Proof. An antipode-preserving $h : S^2 \to S^1$ restricted to the equator would be a nulhomotopic (it extends over a hemisphere) antipode-preserving loop; but an antipode-preserving map $S^1 \to S^1$ induces an odd, hence nonzero, multiple on π_1 (lift and use $h(-x) = -h(x)$ to see that the lift of half the circle ends at a half-odd-integer). ■

Consequence (bisection theorem): two bounded polygonal regions in the plane can be simultaneously bisected by a single line. Also, no subset of $\mathbb{R}^2$ is homeomorphic to S^2.

9.7 Deformation retracts and homotopy type

Lemma 9.10

If $h, k : X \to Y$ are homotopic via H with $\alpha(t) = H(x_0, t)$, then $k_* = \hat{\alpha} \circ h_*$. In particular homotopic maps fixing the base point throughout induce the same homomorphism.

$A \subset X$ is a **deformation retract** if the identity of X is homotopic to a retraction onto A by a homotopy fixing A pointwise; then the inclusion induces an isomorphism on π_1. Examples: S^{n-1} is a deformation retract of $\mathbb{R}^n - 0$; the theta space is a deformation retract of the plane minus two points. Maps $f : X \to Y$, $g : Y \to X$ with $g \circ f \simeq \mathrm{id}_X$, $f \circ g \simeq \mathrm{id}_Y$ make X, Y **homotopy equivalent**, and then f_* is an isomorphism (by the lemma). A space homotopy equivalent to a point is **contractible**.

9.8 The fundamental groups of spheres and some surfaces

Theorem 9.11

If $X = U \cup V$ with U, V open, $U \cap V$ path connected and $x_0 \in U \cap V$, then $\pi_1(X, x_0)$ is generated by the images of $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$. Hence S^n is simply connected for $n \geq 2$.

Proof. Subdivide a loop so each piece lies in U or in V, and connect the subdivision points to x_0 by paths in $U \cap V$; the loop becomes a product of loops each lying in U or in V. For S^n take U, V the sphere minus the north or south pole, each homeomorphic to $\mathbb{R}^n$, with connected intersection when $n \geq 2$. ■

Example 9.2

(a) $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$; hence $\pi_1(T^2) \cong \mathbb{Z} \times \mathbb{Z}$.
(b) The projective plane $P^2 = S^2/(x \sim -x)$ has the 2-fold covering $S^2 \to P^2$ with simply connected total space, so $\pi_1(P^2) \cong \mathbb{Z}/2$.
(c) $\mathbb{R}^2$ and $\mathbb{R}^n$ $(n \neq 2)$ are not homeomorphic ($\mathbb{R}^n$ minus a point is simply connected for $n \geq 3$); S^2, T^2, P^2 are pairwise non-homeomorphic. The figure eight has non-abelian fundamental group (Chapter 11).